G题同步Buck完整原理图 · Rev C

2026-10-26（北京时间）| 基准：Rev B PDF及仓库可编辑工程 | 工程评审稿

交付内容：嘉立创EDA Pro可编辑工程、8页原生记录预览、BOM、逐引脚接线表、变更说明和静态核对结果。原工程中的全部设计文档与PCB保持原记录；新增独立Rev C原理图，未建立Rev C PCB。

驱动：U1由IR2104PbF改为IR2184PbF，保留单路PWM与内部死区。物理脚号改为1 IN、2 /SD、3 COM、4 LO、5 VCC、6 VS、7 HO、8 VB。禁止直接插入IR2104旧板。

保护：电感门限改为标称±2.2 A，独立1kΩ+1nF保护滤波；U9B增加1.10 A输出告警，经J_CTL.6接STM32 PB11/EXTI11。电感过流/过压仍经FAULT_N硬件锁存。

输入：F1之后增加22Ω预充电阻和并联外置SPST旁路开关接口。开关初始断开；预充至少100ms且检查电压后闭合，闭合后才允许ARM/启动。未闭合时禁止带载运行。

效率：QL并联SS56（60V）肖特基，栅极电阻改4.7Ω；5V/12V辅助电源明确要求开关型。D1仍保留SS56；未把反接P-MOS简单等同于理想二极管。

使用边界：IOUT比较器是快速告警，不是输出串联硬限流。输出电容引起的瞬时电流仍可能超过1A；本版不能据此宣称通过“任何瞬间不超过1A”的测试。

验证边界：已核对原生记录中的113个器件、49个网络、296个连接引脚和13个NC；未执行客户端打开测试、EDA ERC、PCB DRC、Rev C动态仿真、固件测试或实测效率。IR2184名义驱动电流不能证明实际栅极边沿为47ns。

后附8页均从交付的.eprj2原生记录绘制，为预览，不是嘉立创EDA客户端导出。红色表示本版新增或修改器件。

关键连线与装配变更

同名网络全局相连；NC不连接；新图纸与旧PCB未同步

预充主通路	J1.1→F1→FUSED→(R_PRE并联外置旁路开关)→IN_PRE→D1.A2→V_SHUNT_IN→RS_IN→VIN
续流肖特基	D_FREE.K1→SW；D_FREE.A2→GND；并联于QL的D/S，布局紧贴QL
电感保护滤波	IL_BI→R_IL_PROT 1kΩ→IL_PROT；C_IL_PROT 1nF→GND；IL_PROT→U5.2和U9.3
正/负电感参考	R9=10k；R10A=49.9k、R10B=100Ω；R15A=49.9k、R15B=100Ω；R16=10k，均0.1%
输出电流告警	IOUT_RAW→R_IOUT_PROT 1kΩ→IOUT_PROT→U9.6；C_IOUT_PROT 1nF→GND
输出参考与中断	3V3_A→20k→VREF_IOUT→10k→GND；100nF旁路；U9.5=VREF_IOUT；U9.7→IOUT_OC_N
控制接口必改	IOUT_OC_N经10k上拉到3V3，J_CTL.6由GND改接PB11/EXTI11；禁止沿用旧排线
ADC与模拟电源	C_IL_ADC：4.7nF→1nF；R_VDDA：10Ω→1Ω；RS_L保持25mΩ，严禁用50mΩ替代
振铃调试预留	SW→R_SN 10Ω→SN_MID→C_SN 1nF/100V→GND；R_SN与C_SN默认DNP，实测后再装
LCD装配	仅对附件所述3.3V LCD模块：短接模块内部J1焊桥；它不是主板输入端子J1。背光电阻不在主板内改动
软件配套	30kHz起调，25 – 30kHz范围复核；新增PB11告警处理、连续短路检测与锁存、重新ARM、标定和预充流程
仍需验证	输出瞬时硬限流、整机≥85%效率、器件数据手册与实物封装、开关波形、短路响应、原生ERC/PCB DRC

R_PRE需选实际功率/脉冲电阻与封装，工程中不绑定0805封装；本版没有可用于生产的Rev C PCB。

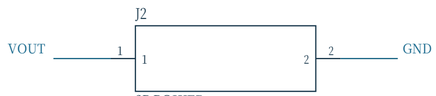
01 主功率级 / 15 – 25V → 10V, 1A

Rev C 原理图修订 / 未建Rev C PCB / 未运行客户端ERC、DRC

Rev C：F1→预充并联旁路→D1；D_FREE阳极GND、阴极SW；D1仍为SS56。

2P POWER

TERM_5.08_2P [TBD]

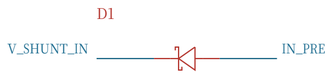


2P POWER

TERM_5.08_2P [TBD]

2A / 5x20

FUSE_5X20_P22.5_TBD [TBD]



SS56

SMB_DO214AA

RS_IN

50mΩ / 1% / >=1W

R2512

RS_OUT

50mΩ / 1% / >=1W

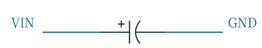
R2512

RS_L

25mΩ / 1% / >=1W

R2512

C1



470uF / 50V

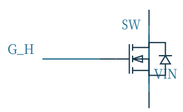
CP_D12.5_P5_TBD [TBD]

C2

1uF / 50V

C1206

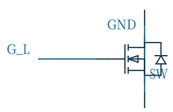
QH



IRF540N

TO220_3

QL



IRF540N

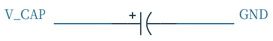
TO220_3

L1

220uH / Isat>=3A

L_TH_D18_P10_TBD [TBD]

C3



470uF / 35V

CP_D12.5_P5_TBD [TBD]

C4

10uF / 35V

C1206

R_BL

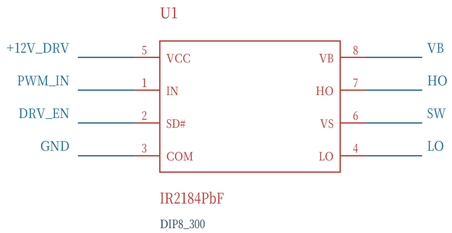
2.2kΩ / 0.25W

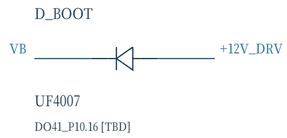
R1206

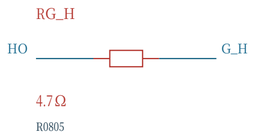
02 IR2184 单路PWM驱动与开关辅助电源

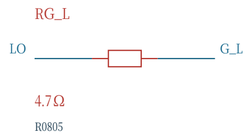
Rev C 原理图修订 / 未建Rev C PCB / 未运行客户端ERC、DRC

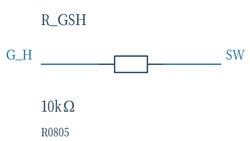
U1改IR2184PbF；物理脚号已重排，严禁直接插入IR2104旧板；单PWM，内部死区。

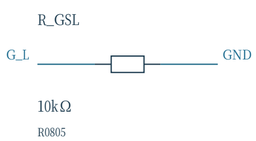


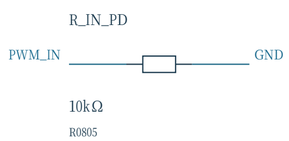


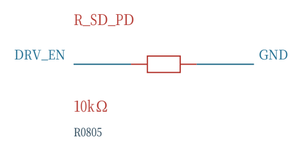


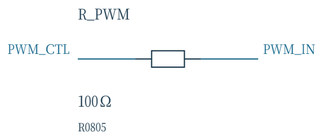


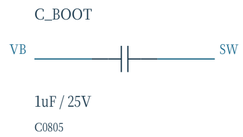


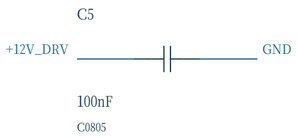


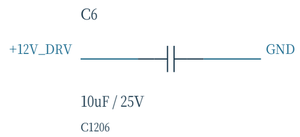


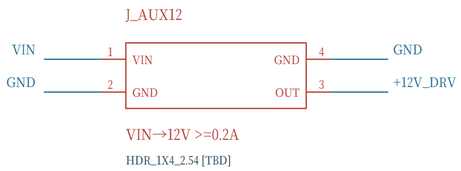


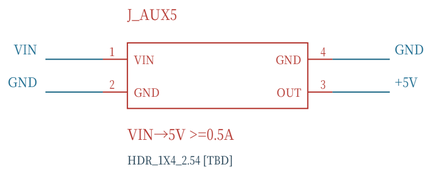








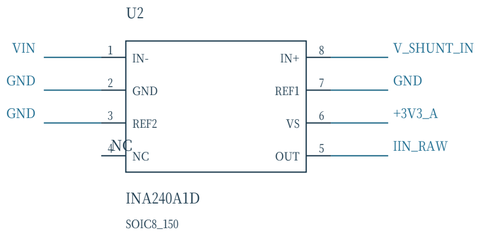


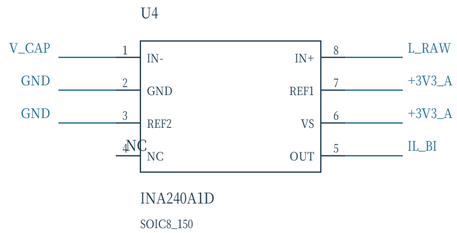


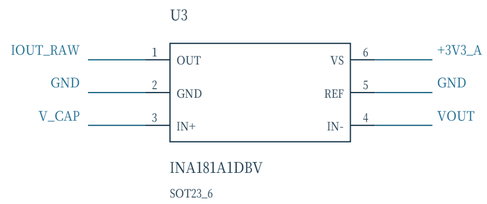
03 电流采样与模拟电源

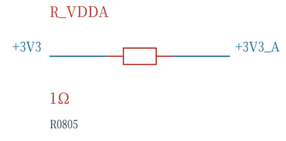
Rev C 原理图修订 / 未建Rev C PCB / 未运行客户端ERC、DRC

RS_L严格25mΩ；R_VDDA=1Ω；U4零点=3V3_A/2，电流增益0.5V/A。





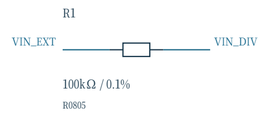


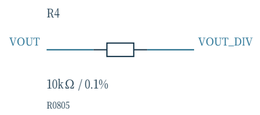


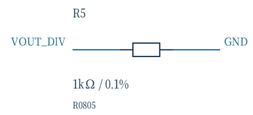
04 电压分压、五路ADC滤波与钳位

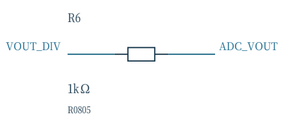
Rev C 原理图修订 / 未建Rev C PCB / 未运行客户端ERC、DRC

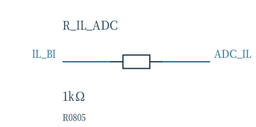
C_IL_ADC=1nF；比较器走独立IL_PROT/IOUT_PROT，不使用ADC滤波节点。

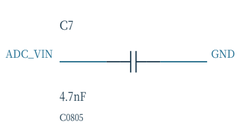


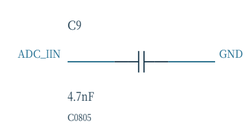


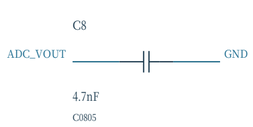


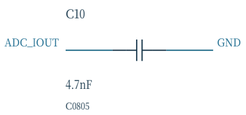


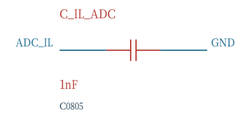


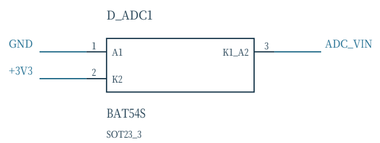


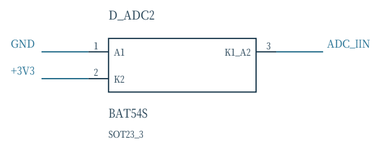


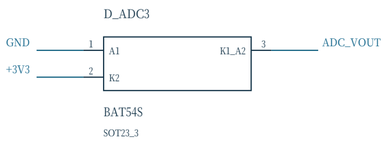


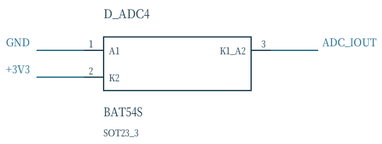


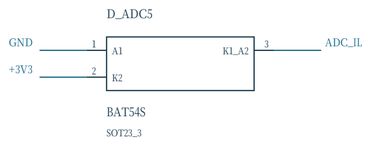








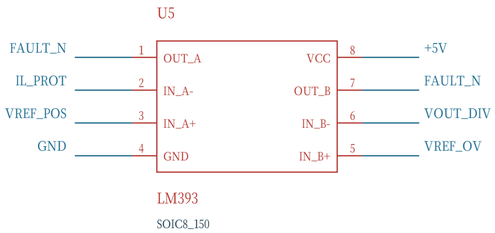


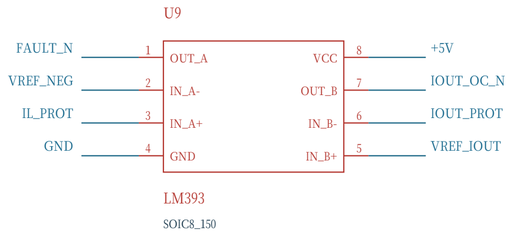


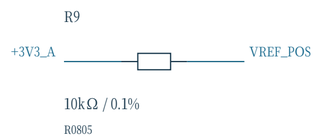
05 ±2.2A电感保护、1.10A输出告警及过压锁存

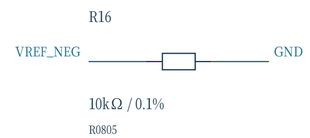
Rev C 原理图修订 / 未建Rev C PCB / 未运行客户端ERC、DRC

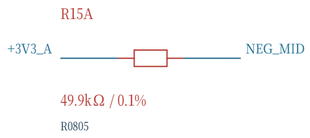
IL阈值2.75V/0.55V≈±2.2A；U9B新增1.10A输出告警→PB11；IL/OVP拉低FAULT_N。

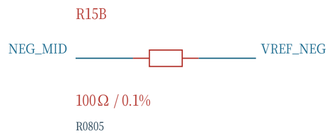


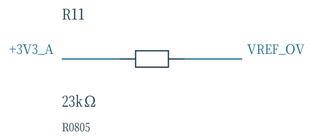


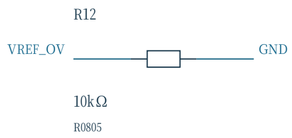


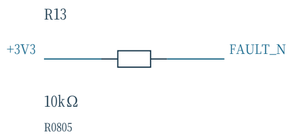


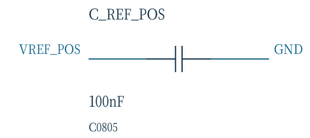


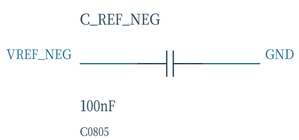


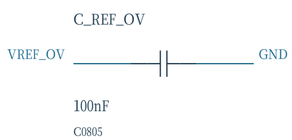


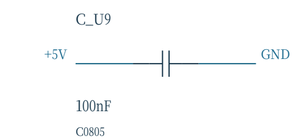










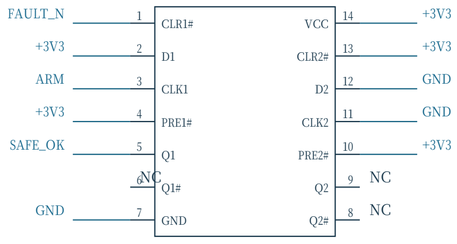


06 上电复位、故障锁存与使能

Rev C 原理图修订 / 未建Rev C PCB / 未运行客户端ERC、DRC

原故障锁存及ARM时序保留；清故障后重新ARM；MCU_EN=0才关闭两路。

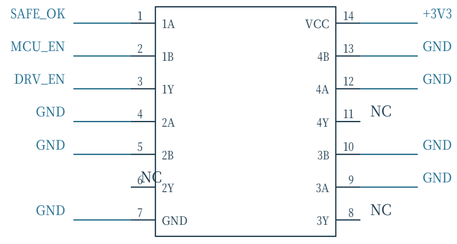
U6



SN74HC74DR

SOIC14_150

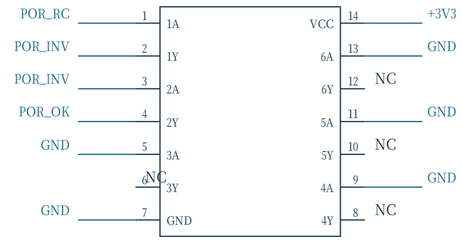
U7



SN74HC08DR

SOIC14_150

U8



SN74HC14DR

SOIC14_150

R14

10kΩ

R0805

R_EN_PD

10kΩ

R0805

R_ARM_PD

10kΩ

R0805

C11

100nF

C0805

C_U6

100nF

C0805

C_U7

100nF

C0805

C_U8

100nF

C0805

D_POR

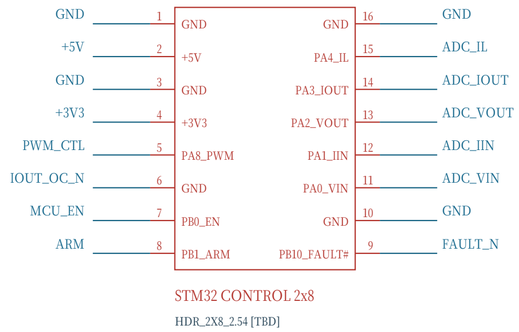
BAT54

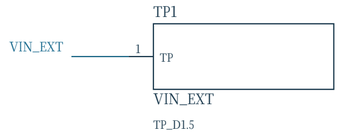
SOT23_3

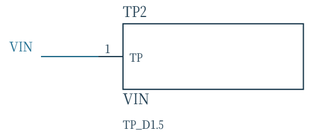
07 STM32控制板接口与测试点

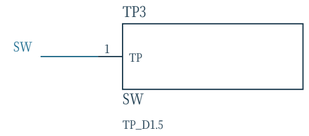
Rev C 原理图修订 / 未建Rev C PCB / 未运行客户端ERC、DRC

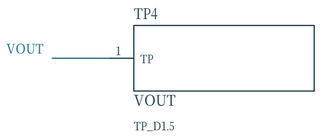
J_CTL.6由GND改IOUT_OC1N→PB11/EXTI11；LCD模块J1焊桥短接；辅助电源必须开关型。

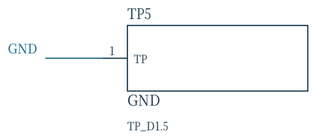


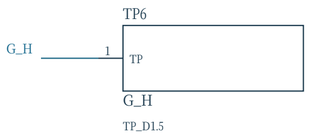


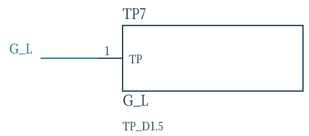


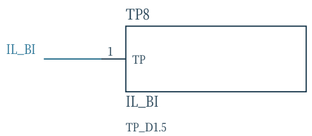


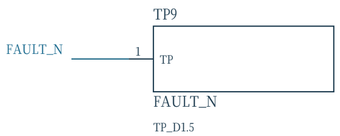


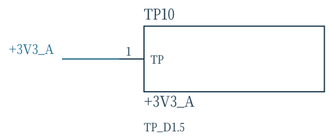












08 预充、续流、保护滤波与新增测试点

Rev C 原理图修订 / 未建Rev C PCB / 未运行客户端ERC、DRC

J_BYP接外置常开SPST旁路开关≥30VDC/3A；上电预充后闭合，停机断电后打开。

R_PRE
22Ω / 5W / pulse≥0.3J

J_BYP
SPST旁路接口 ≥30VDC/3A

D_FREE
SS56 / 60V / 5A

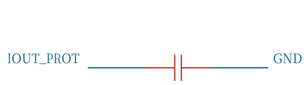
R_SN
10Ω / 0.25W / DNP

C_SN
1nF / 100V C0G / DNP

R_IL_PROT
1kΩ / 1%

C_IL_PROT
1nF / C0G

R_IOUT_PROT
1kΩ / 1%



C_IOUT_PROT
1nF / C0G

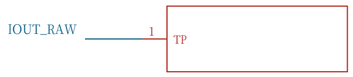
R_IOUT_TOP
20kΩ / 0.1%

R_IOUT_BOT
10kΩ / 0.1%

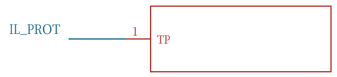
C_REF_IOUT
100nF

R_IOUT_PU
10kΩ

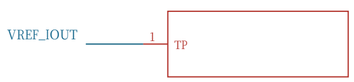
C_DRV
1uF / 25V X7R



TP11
IOUT_RAW



TP12
IL_PROT



TP13
VREF_IOUT